

LAWS OF MOTION MCQ

1. Motion is best described as change in position of object with respect to:

- a) absolute space
- b) surroundings in given time
- c) velocity only
- d) acceleration

Correct answer: b

2. Displacement differs from distance because it is:

- a) scalar quantity
- b) vector with direction magnitude
- c) always positive
- d) path length

Correct answer: b

3. Velocity defined as vector measurement includes:

- a) speed only
- b) direction rate of motion
- c) distance time
- d) acceleration change

Correct answer: b

4. Acceleration rate of change of velocity respects:

- a) scalar nature
- b) both magnitude direction
- c) time only
- d) distance covered

Correct answer: b

5. Dynamics branch of mechanics considers:

- a) rest bodies
- b) moving bodies under forces
- c) no forces
- d) equilibrium only

Correct answer: b

6. Kinematics studies motion without:

- a) velocity consideration
- b) cause of motion
- c) acceleration
- d) displacement?

Correct answer: b

7. Aristotle classified natural motion as force:

- a) dependent
- b) independent like falling objects
- c) violent only
- d) circular?

Correct answer: b

8. Galileo's key concept states natural state of bodies is:

- a) rest only
- b) rest or uniform motion
- c) accelerated motion
- d) circular motion?

Correct answer: b

9. Inertia defined as inability of body to change state by:

- a) external force

b) itself alone

c) gravity

d) friction?

Correct answer: b

10. Inertia depends mainly upon body's:

a) velocity

b) mass

c) shape

d) colour?

Correct answer: b

11. Inertia of rest example when bus starts passenger falls:

a) forward

b) backward

c) sideways

d) upward?

Correct answer: b

12. Coin remains at rest due to inertia when cardboard is:

a) pulled quickly away

b) pushed slowly

c) stationary

d) rotated?

Correct answer: a

13. Inertia of direction evident when car turns sharply driver leans:

a) forward

b) sideways

c) backward

d) upward?

Correct answer: b

14. Linear momentum product of mass velocity acts in direction of:

a) force

b) acceleration

c) velocity

d) displacement?

Correct answer: c

15. Newton's First Law states body at rest or uniform motion remains unless acted by:

a) gravity

b) force

c) friction

d) inertia?

Correct answer: b

16. Force defined as push pull changing object's:

a) position shape speed direction

b) colour only

c) mass

d) volume?

Correct answer: a

17. Thrust means force acting perpendicularly on surface:

a) area

b) parallel

c) tangentially

d) obliquely?

Correct answer: a

18. Like parallel forces act along:

a) opposite directions

b) same direction

c) perpendicular

d) random?

Correct answer: b

19. Equilibrant force equals resultant but:

a) same direction

b) opposite direction

c) half magnitude

d) double?

Correct answer: b

20. Moment of force measured by product force perpendicular:

a) velocity

b) distance from axis

c) mass

d) time?

Correct answer: b

21. Torque SI unit is:

a) joule

b) newton metre

c) dyne

d) kg m/s?

Correct answer: b

22. Moment of couple product of force perpendicular distance between:

- a) lines of action
- b) masses
- c) velocities
- d) accelerations?

Correct answer: a

23. Principle of moments states algebraic sum moments about axis zero for:

- a) motion
- b) equilibrium
- c) acceleration
- d) rotation?

Correct answer: b

24. Newton's Second Law rate of change momentum proportional to applied:

- a) velocity
- b) force
- c) mass
- d) distance?

Correct answer: b

25. $F = m a$ derives from second law where a change velocity:

- a) per distance
- b) per time
- c) squared
- d) inversely?

Correct answer: b

26.1 Newton defined as force producing acceleration 1 m/s^2 on mass:

- a) 1 mg
- b) 1 g
- c) 1 kg
- d) 1 tonne?

Correct answer: c

27. Impulse equals force time also change in:

- a) kinetic energy
- b) momentum
- c) velocity
- d) acceleration?

Correct answer: b

28. Newton's Third Law action reaction equal opposite act on:

- a) same body
- b) different bodies
- c) no bodies
- d) medium?

Correct answer: b

29. Conservation linear momentum holds if no external force acts total momentum:

- a) increases
- b) decreases
- c) conserved
- d) zero?

Correct answer: c

30. Rocket propulsion based on third law gases ejected backward rocket moves:

- a) backward
- b) forward
- c) sideways
- d) upward only?

Correct answer: b

31. Escape velocity sufficient to overcome earth's:

- a) magnetic field
- b) gravitational pull
- c) atmosphere
- d) rotation?

Correct answer: b

32. Universal gravitation every particle attracts others proportional product masses inversely:

- a) distance
- b) square distance
- c) cube distance
- d) root distance?

Correct answer: b

33. G universal constant value $6.674 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$ units:

- a) N kg
- b) $\text{Nm}^2 \text{ kg}^{-2}$
- c) m/s^2
- d) J/kg?

Correct answer: b

34. Acceleration due gravity g average earth surface:

- a) 8.9 m/s^2
- b) 9.8 m/s^2
- c) 10 m/s^2
- d) 9.0 m/s^2 ?

Correct answer: b

35. g maximum at earth's:

- a) equator
- b) poles
- c) centre
- d) height?

Correct answer: b

36. Mass measures amount of:

- a) weight
- b) matter
- c) volume
- d) density?

Correct answer: b

37. Weight mg vector towards earth's:

- a) surface
- b) centre
- c) poles
- d) equator?

Correct answer: b

38. Apparent weight in accelerating lift upward $R = m(g + a)$ where a :

- a) downward

b) upward acceleration

c) zero

d) negative?

Correct answer: b

39. Weightlessness when apparent weight:

a) maximum

b) zero free fall

c) double

d) half?

Correct answer: b

40. Astronauts weightless because in free fall orbital:

a) zero gravity

b) velocity

c) station

d) space vacuum?

Correct answer: b

41. Distance scalar measures length without:

a) time

b) direction

c) velocity

d) acceleration?

Correct answer: b

42. Uniform motion straight line constant:

a) acceleration

b) velocity

c) distance

d) displacement?

Correct answer: b

43. Statics deals with bodies:

a) moving

b) rest under forces

c) accelerating

d) rotating?

Correct answer: b

44. Kinetics considers the cause of:

a) rest

b) motion forces

c) equilibrium

d) inertia?

Correct answer: b

45. Aristotle violent motion caused by external:

a) natural agent

b) accidental agent

c) gravity

d) inertia?

Correct answer: b

46. Galileo proved different masses fall at the same rate in:

a) air

b) vacuum

c) water

d) oil?

Correct answer: b

47. When car stops suddenly passengers fall:

- a) backward
- b) forward inertia motion
- c) sideways
- d) down?

Correct answer: b

48. Athlete runs before jump utilizes inertia of:

- a) rest
- b) motion
- c) direction
- d) mass?

Correct answer: b

49. Roller coaster loop passengers feel thrown outward due inertia:

- a) rest
- b) motion
- c) direction
- d) gravity?

Correct answer: c

50. Linear momentum vector quantity direction same as:

- a) force
- b) acceleration
- c) velocity
- d) displacement?

Correct answer: c

51. Balanced forces resultant:

- a) non-zero

b) zero equilibrium

c) maximum

d) minimum?

Correct answer: b

52. Unbalanced forces cause change in:

a) rest

b) motion

c) shape only

d) colour?

Correct answer: b

53. Turning effect force about pivot maximum when perpendicular distance:

a) minimum

b) maximum

c) zero

d) negative?

Correct answer: b

54. Door swings about:

a) knob

b) hinge axis

c) centre

d) floor?

Correct answer: b

55. Spanner tightens nut better longer handle because greater:

a) force

b) moment

c) mass

d) velocity?

Correct answer: b

56. Seesaw balances when moments both sides:

a) unequal

b) equal

c) maximum

d) minimum?

Correct answer: b

57. Gears change speed by altering:

a) force

b) torque

c) mass

d) direction only?

Correct answer: b

58. Bicycle pedal torque increases with longer:

a) wheel

b) crank arm

c) chain

d) seat?

Correct answer: b

59. Steering wheel small transfers torque with:

a) more effort

b) less effort

c) same effort

d) no effort?

Correct answer: b

60. Change momentum $\Delta p = m(v - u)$ second law $F = \Delta p / \Delta t$:

- a) distance
- b) time
- c) mass
- d) force?

Correct answer: b

61. Centripetal acceleration directed towards circle:

- a) tangent
- b) centre
- c) outside
- d) chord?

Correct answer: b

62. 1 dyne force accelerates 1 g by:

- a) 1 m/s^2
- b) 1 cm/s^2
- c) 9.8 cm/s^2
- d) 1 km/s^2 ?

Correct answer: b

63. Impulse $J = Ft$ unit:

- a) joule
- b) newton second
- c) kg m/s
- d) both b c?

Correct answer: d

64. Cricket player moves hands backward catching fast ball reduce:

- a) force

b) impulse same change momentum

c) velocity

d) mass?

Correct answer: b

65. Action reaction third law always on different:

a) directions only

b) bodies

c) times

d) mediums?

Correct answer: b

66. Two bodies collide momentum conserved no external:

a) friction

b) force

c) gravity

d) tension?

Correct answer: b

67. Rocket fuel burnt gases high speed ejected produces reaction force:

a) backward

b) forward rocket

c) sideways

d) downward?

Correct answer: b

68. Rocket mass decreases velocity:

a) decreases

b) increases conservation momentum

c) constant

d) zero?

Correct answer: b

69. Orbital velocity less than escape velocity:

a) escapes

b) returns earth

c) orbits stable

d) stops?

Correct answer: b

70. Gravitational force attractive independent of:

a) distance

b) medium

c) mass

d) size?

Correct answer: b

71. Earth radius approx for g calculation:

a) 6378 km

b) 6400 km

c) 6000 km

d) 7000 km?

Correct answer: a

72. Earth mass calculated $M = \frac{g R^2}{G}$:

a) G

b) m

c) r

d) h?

Correct answer: a

73. g decreases with altitude because r :

- a) decreases
- b) increases
- c) constant
- d) zero?

Correct answer: b

74. At earth centre g :

- a) maximum
- b) minimum zero
- c) 9.8
- d) infinite?

Correct answer: b

75. Weight vector direction always towards earth:

- a) surface
- b) centre
- c) poles
- d) equator?

Correct answer: b

76. Mass constant unlike weight varies with:

- a) location g
- b) size
- c) shape
- d) temperature?

Correct answer: a

77. Lift accelerating downward apparent weight $R = m(g - a)$:

- a) increases

b) decreases

c) same

d) zero?

Correct answer: b

78. Free fall $a = g$ apparent weight:

a) mg

b) zero

c) $2mg$

d) $mg/2$?

Correct answer: b

79. Roller coaster bottom passengers feel heavier apparent weight:

a) less

b) more $g + a$

c) same

d) zero?

Correct answer: b

80. Astronauts fall freely around earth orbital velocity causes:

a) gravity absent

b) weightlessness

c) high weight

d) equilibrium?

Correct answer: b

81. Universal law helps predict:

a) weather

b) tides moon pull

c) rain

d) wind?

Correct answer: b

82. Inertia stone greater than football same size because:

a) less mass

b) more mass

c) shape

d) colour?

Correct answer: b

83. Unlike parallel forces equal opposite cause:

a) motion

b) equilibrium

c) acceleration

d) rotation?

Correct answer: b

84. Resultant force zero body:

a) accelerates

b) equilibrium

c) rotates

d) vibrates?

Correct answer: b

85. Torque vector perpendicular to plane force:

a) velocity

b) distance

c) both b

d) mass?

Correct answer: c

86.Clockwise moment convention:

- a) positive
- b) negative
- c) zero
- d) infinite?

Correct answer: b

87.Derivation second law assumes straight line initial final:

- a) acceleration
- b) velocity
- c) momentum
- d) force?

Correct answer: b

88.Unit force accelerates 1 kg by 1 m/s² called:

- a) dyne
- b) newton
- c) kgf
- d) gf?

Correct answer: b

89.kgf equals approx:

- a) 1 N
- b) 9.8 N
- c) 10 N
- d) 100 N?

Correct answer: b

90.Proof conservation momentum uses $F_A = -F_B$ third:

- a) first law

b) second law

c) third law

d) gravitation?

Correct answer: c

91. Rocket nozzle ejects gas high speed backward reaction propels:

a) backward

b) forward

c) down

d) up only?

Correct answer: b

92. Escape velocity greater than orbital for:

a) orbit

b) escape gravity

c) return

d) stop?

Correct answer: b

93. G depends not on medium unlike some:

a) electric forces

b) magnetic

c) gravitational constant

d) nuclear?

Correct answer: c

94. g poles greater equator because radius:

a) greater poles

b) smaller poles

c) same

d) rotation?

Correct answer: b

95. Weight moon less earth because g smaller about:

a) $1/6$

b) $1/2$

c) same

d) double?

Correct answer: a

96. Apparent weight elevator rest:

a) $mg + a$

b) mg

c) $mg - a$

d) zero?

Correct answer: b

97. Space station free fall astronauts:

a) weightless relative acceleration

b) heavy

c) normal

d) floating mass zero?

Correct answer: a

98. Inertia rest gold coin table cloth pulled quickly:

a) moves cloth

b) stays coin

c) falls

d) rotates?

Correct answer: b

99. Bicycle turns rider leans prevent falling inertia:

- a) rest
- b) motion
- c) direction
- d) mass?

Correct answer: c

100. Conservation momentum gun recoils bullet fired equal opposite:

- a) mass
- b) momentum
- c) velocity
- d) energy?

Correct answer: b